

Problem 1

- (a) Let $\sigma \in S_n$, write $\sigma = a_1 b_1 \dots a_m b_m$ where a_i, b_i are transpositions

Consider $a_k b_k$

$$a_k b_k = (ab)(cd) \text{ s.t. } a, b, c, d \in \{1, 2, \dots, n\}$$

$$(ab)(cd) = (acb)(acd)$$

Thus since we have an even amount of transpositions we can write them all as 3-cycles.

- (b) Let (abc) and $(defg)$ be two 3 cycles in A_n

$$\text{Consider } (abc)(defg)(abc)^{-1} = (abc)(defg)(acb) = (defg)$$

- (c) i. Let $N \trianglelefteq A_n$ if $N = \{e\}$ then done so assume $N \neq \{e\}$

Let $e \neq \sigma \in N$ s.t. it has the most fixed points.

- ii. Let $\sigma = (a_1 b_1)(a_2 b_2) \dots (a_t b_t)$ bet disjoint transpositions.

t is not 1 since this would mean σ is odd so $t \geq 2$

Since these are distinct this means we fix $n - 2t \geq n - 4$ elements, and since $n \geq 5 \implies$

$n - 2t \geq n - 4 \geq 1$ thus there is at least one element that is fixed by σ

Let $r \in \{1, \dots, n\}$ s.t. r is fixed in σ

Let $\tau = (a_1 b_1 r)$

Consider $\tau \sigma \tau^{-1} \sigma^{-1}$

Since N is normal we know $\tau \sigma \tau^{-1} \in N$ and since $\sigma^{-1} \in N$ we know $[\tau, \sigma] \in N$

$$= \tau(a_1 b_1) \tau^{-1} \tau(a_2 b_2) \tau^{-1} \dots \tau(a_t b_t) \tau^{-1} \sigma$$

$$= (b_1 r)(a_2 b_2) \dots (a_t b_t) \sigma$$

$$= (b_1 r)(a_1 b_1) = (a_1 r b_1)$$

Thus $[\tau, \sigma]$ fixes $n - 3$ points

$n - 2t \leq n - 4 < n - 3$ thus the commutator fixes more points and is not the identity, thus σ cannot be written as disjoint transpositions.

- iii. Since 3-cycles cant be written as two disjoint transpositions if σ is a 3 cycle then we are done so assume not

The only way for a permutation to move an even amount of elements if it can be written as disjoint transpositions thus it σ cannot move only 4 elements then it could be written as 2 disjoint transpositions thus σ must move at least 5 elements.

- iv. Thus $\sigma = (a_1 a_2 a_3 \dots)(\dots) \dots (\dots)$

Let r, s be two elements not in $\{a_1, a_2, a_3\}$ that are moved by σ .

Let $\tau = (a_3 r s)$

Consider $\tau \sigma \tau^{-1} \sigma^{-1}$

Since N is normal we know $\tau \sigma \tau^{-1} \in N$ and since $\sigma^{-1} \in N$ we know $[\tau, \sigma] \in N$

Let x be fixed by σ then $\tau \sigma \tau^{-1} \sigma^{-1}(x) = \tau \sigma \tau^{-1}(x)$ and since τ only touches things that

are moved by σ thus $\tau\sigma(x) = \tau(x) = x$

Thus if x is fixed by σ it is fixed by $[\tau, \sigma]$

Consider a_2 , σ moves a_2 now consider $\tau\sigma\tau^{-1}\sigma^{-1}(a_2) = \tau\sigma\tau^{-1}(a_1) = \tau\sigma(a_1) = \tau(a_2) = a_2$

Thus the commutator fixes a_2 thus this is a contradiction thus σ must be a 3-cycle

Thus for all 3 cycles $x \in A_n$ we have that $\sigma x \sigma^{-1}$ is also a 3 cycle and in N since N is normal.

Thus let $\sigma = (abc)$ and let $x = (dfg)$

$(abc)(dfg)(acb) = (dfg)$ thus we have all 3 cycles in N and since A_n is generated by 3-cycles thus $N = A_n$

Problem 2

$|S_p| = p!$ thus the order of a p -Sylow is just p

Thus the number of distinct p cycles is $(p-1)!$ thus there are $(p-1)!$ groups generated by each of these p cycles

However, proven in last homework since in these groups are of order p we know it has $p-1$ generators thus we are overcounting by a factor of $p-1$ since we count each group $p-1$ times thus

$$n_p = \frac{(p-1)!}{p-1} = (p-2)!$$

Thus by Sylow 3 we know $n_p = (p-2)! \equiv 1 \pmod{p} \implies (p-1)! \equiv (p-1) \pmod{p} \implies (p-1)! \equiv -1 \pmod{p}$

Problem 3

Consider $SL_2(F_5) = \{2 \times 2 \text{ matrices} \mid \det = 1\}$

Thus for matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ in $SL_2(F_5)$ we know $ad - bc = 1$

Consider the first column we have $5^2 - 1$ choices since all it has to be is non zero

Then for the second column we have 5 choices for b but then d is fixed to satisfy $ad - bc = 1$ thus we have 120 choices thus $|SL_2(F_5)| = 120$

Thus since $PSL_2(F_5) = \frac{SL_2(F_5)}{\{\pm I\}}$ we know $|PSL_2(F_5)| = \frac{|SL_2(F_5)|}{|\{\pm I\}|} = 60$

Let x and y be two classes of matrices

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix}$$

Consider A

$A^5 = I$ thus $\langle AI \rangle = \langle A \rangle$ thus $\langle A \rangle$ is a subgroup of order 5

Consider B

$B^5 = I$ thus $\langle BI \rangle = \langle B \rangle$ thus $\langle B \rangle$ is a subgroup of order 5

Since $A \neq B$ and $-A \neq B$ we know that these are distinct in $PSL_2(F_5)$

Thus since a group of order 60 has more than 1 subgroup of order 5 we know it must have 5 by Sylow 3, and thus it must be isomorphic to A_5

Problem 4

- (a) Consider $SL_2(F_7) = \{2 \times 2 \text{ matrices} \mid \det = 1\}$

Thus for matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ in $SL_2(F_7)$ we know $ad - bc = 1$

Consider the first column we have $7^2 - 1$ choices since all it has to be is non zero

Then for the second column we have 7 choices for b but then d is fixed to satisfy $ad - bc = 1$ thus we have 120 choices thus $|SL_2(F_7)| = 7(7^2 - 1) = 336$

Thus since $PSL_2(F_7) = \frac{SL_2(F_7)}{\{\pm I\}}$ we know $|PSL_2(F_7)| = \frac{|SL_2(F_7)|}{|\{\pm I\}|} = 168$

- (b) Let x and y be two classes of matrices

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

Consider A

$A^7 = I$ thus $\langle AI \rangle = \langle A \rangle$ thus $\langle A \rangle$ is a subgroup of order 7

Consider B

$B^7 = I$ thus $\langle BI \rangle = \langle B \rangle$ thus $\langle B \rangle$ is a subgroup of order 7

Since $A \neq B$ and $-A \neq B$ we know that these are distinct in $PSL_2(F_7)$

Thus since a group of order 168 has more than 1 subgroup of order 7 we know it must have 8 by Sylow 3.

- (c) Let $N \trianglelefteq G$ s.t. $[G : N] = 2$ this means that $|N| = 84$ thus $n_7 = 1$ by Sylow 3. However, this is a contradiction since proven in class if $P \trianglelefteq N \trianglelefteq G$ and P is a p -Sylow then $P \trianglelefteq G$ this implies that the one normal seven Sylow is normal in G , but this is a contradiction of Sylow 2 thus it can't be index 2.

- (d) Suppose $[G : N] = 3$ this means $|N| = 56$ thus $n_7 = 1, 8$ by Sylow 3. It must have 8 by shown above thus it has all 7-Sylow subgroups of G . Thus, it must contain both H and K

Thus by group closure it contains element $u = xy^2x$ s.t. $H = \langle x \rangle$ and $K = \langle y \rangle$

$$u = AB^2A$$

$$u^2 = xy^2xxy^2x = xy^2x^2y^2x$$

$$u^3 = xy^2x^2y^2xxy^2x = xy^2x^2y^2x^2y^2x = AB^2A^2B^2A^2B^2A = I$$

This is a contradiction since N is a group of order 56 and 3 does not divide 56 thus it cannot have a subgroup of order 3 thus $[G : N] \neq 3$

- (e) Assume $[G : N] \neq 6$ thus $|N| = 28$ thus by Sylow 3 $n_7 = 1$ this is a contradiction by similar proof above thus $[G : N] \neq 6$

- (f) Assume G is not simple and let $N \trianglelefteq G$ be a maximal proper normal subgroup this means that G/N must be simple thus we only have to check for groups of prime order that divide 168 and since 60 does not divide 168 we don't have to check this
Thus $[G : N] = 7$ since we checked for 2,3 thus $|N| = 24$
- (g) Thus by Sylow 3 we know that $n_2(N) = 1, 3$
Assume $n_2(N) = 1$ then we know there is a normal 2-Sylow H of N that is normal in G . Thus, G/H is a subgroup of order 21 since $168/8 = 21$ thus by Sylow 3 this has a normal 7-Sylow $\bar{K}$. Thus, by proven in homework we know corresponds to a normal subgroup in G called K of G thus since $|K/\text{diagup}H| = \bar{K}$ we know that $|K| = 8 \times 7 = 56$
Thus $[G : K] = 3$ which is a contradiction thus $n_2(N) = 3$
- (h) Thus G acts on the 3 groups 2-Sylow groups by conjugation thus there exists a nontrivial homomorphism $\varphi : G \rightarrow S_3$. This map cannot be injective since $|G| > |S_3|$ thus the kernel is nontrivial, thus $\ker(\varphi) = K$ is a proper normal subgroup of G .
By the first isomorphism theorem we know that $G/K \cong \text{Im}(\varphi) = I$ and the $|I| \mid 6$ and is not 1 thus we get $|I| = 3, 6$ thus $[G : K] = |I| = 3, 6$ and above we have shown these are contradictions thus G must be simple